

2. If ...

[Hengfeng Wei \(魏恒峰\)](#)
hfwei@hnu.edu.cn

Sep. 20, 2024

Review

Variables (变量) Data Types (数据类型)

Operators (运算符) Expressions (表达式)

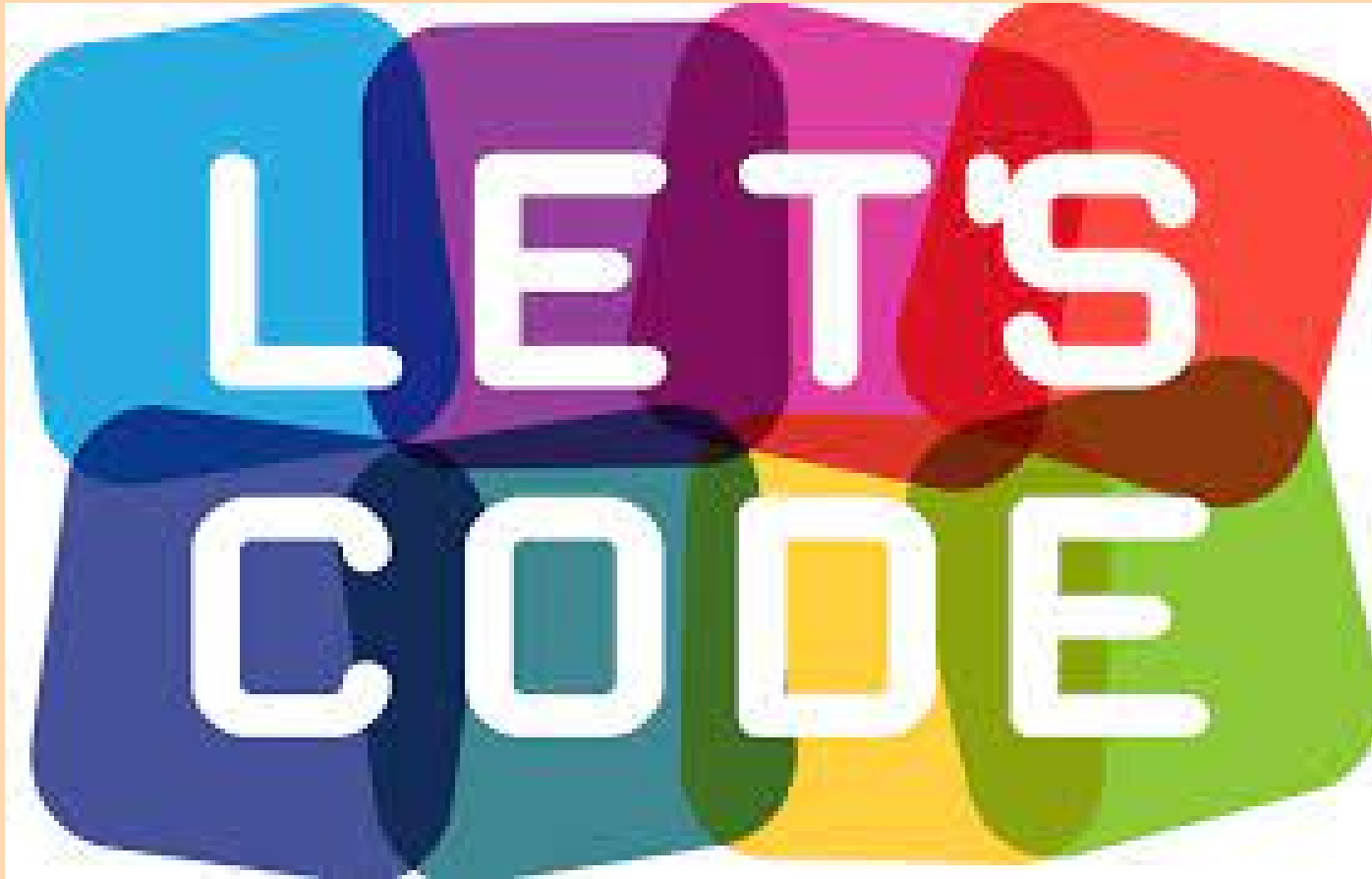
Assignment Statements (赋值语句)

I/O (Input/Output; 输入输出)

Overview

If Statement (`if` 语句)

Switch Statement (`switch` 语句)



min.cpp adult.cpp leap.cpp next-day.cpp

Min of Two

Given two integers a and b , to compute their minimum.

$$\textit{min} = \min\{a, b\}$$

```
min = a >= b ? b : a;
```

(三目运算符)

Do Not Use it Too Much!

Min of Three

Given three integers a , b , and c , to compute their minimum.

$$\textit{min} = \min\{a, b, c\}$$

If with Initializer (C++17)

Input age; check if adult (≥ 18).

adult-outside.cpp adult-inside.cpp

adult-if-init.cpp

Variable Scope: Three Ways

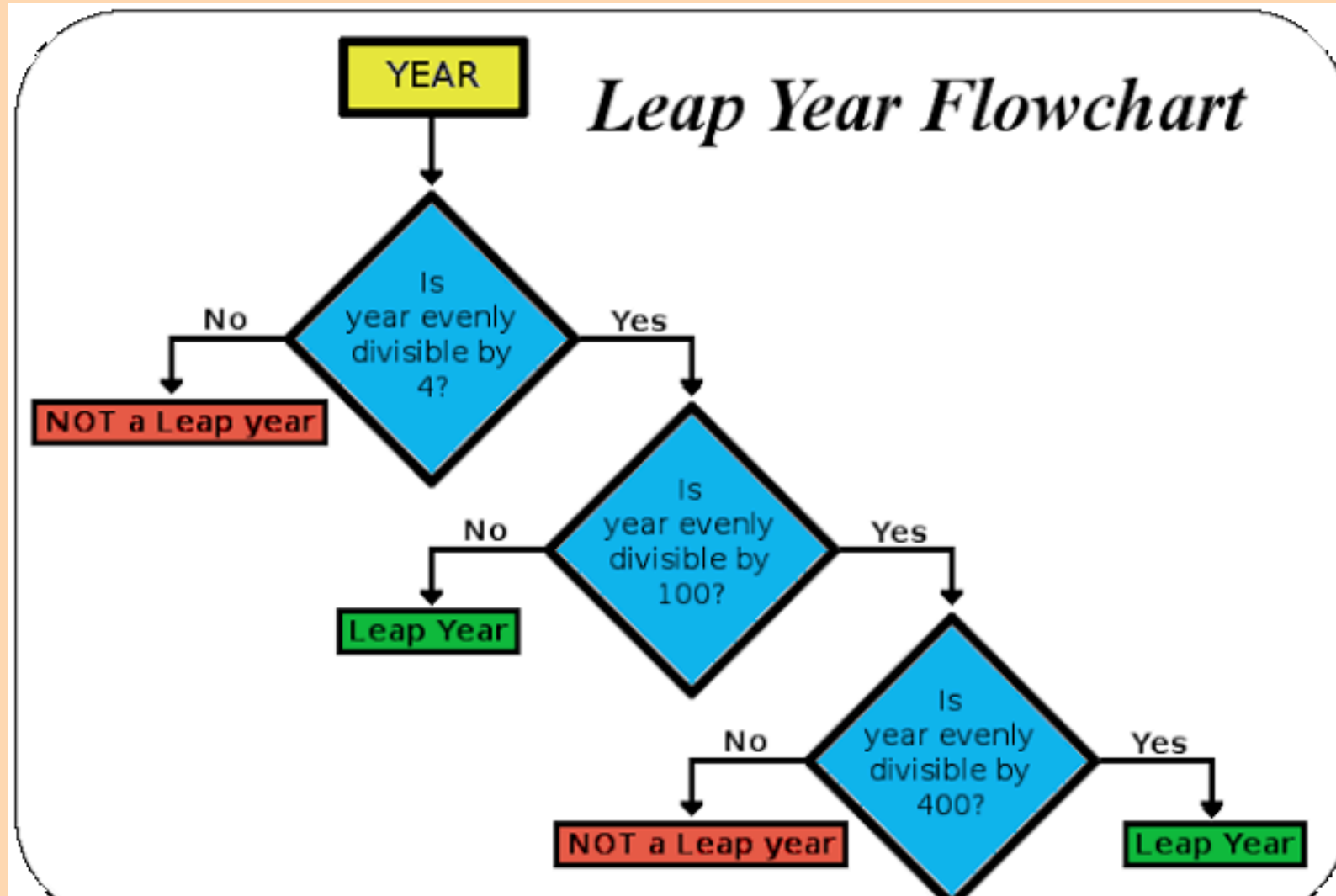
写法	作用域	泄漏?	重复代码?
外部声明	整个函数	是	否
分支内声明	单个分支	否	是
if 初始化器	整个 if-else	否	否

Keep scopes small (CG ES.5)

Leap Year



Leap Year (1): Nested `if/else` (YES)

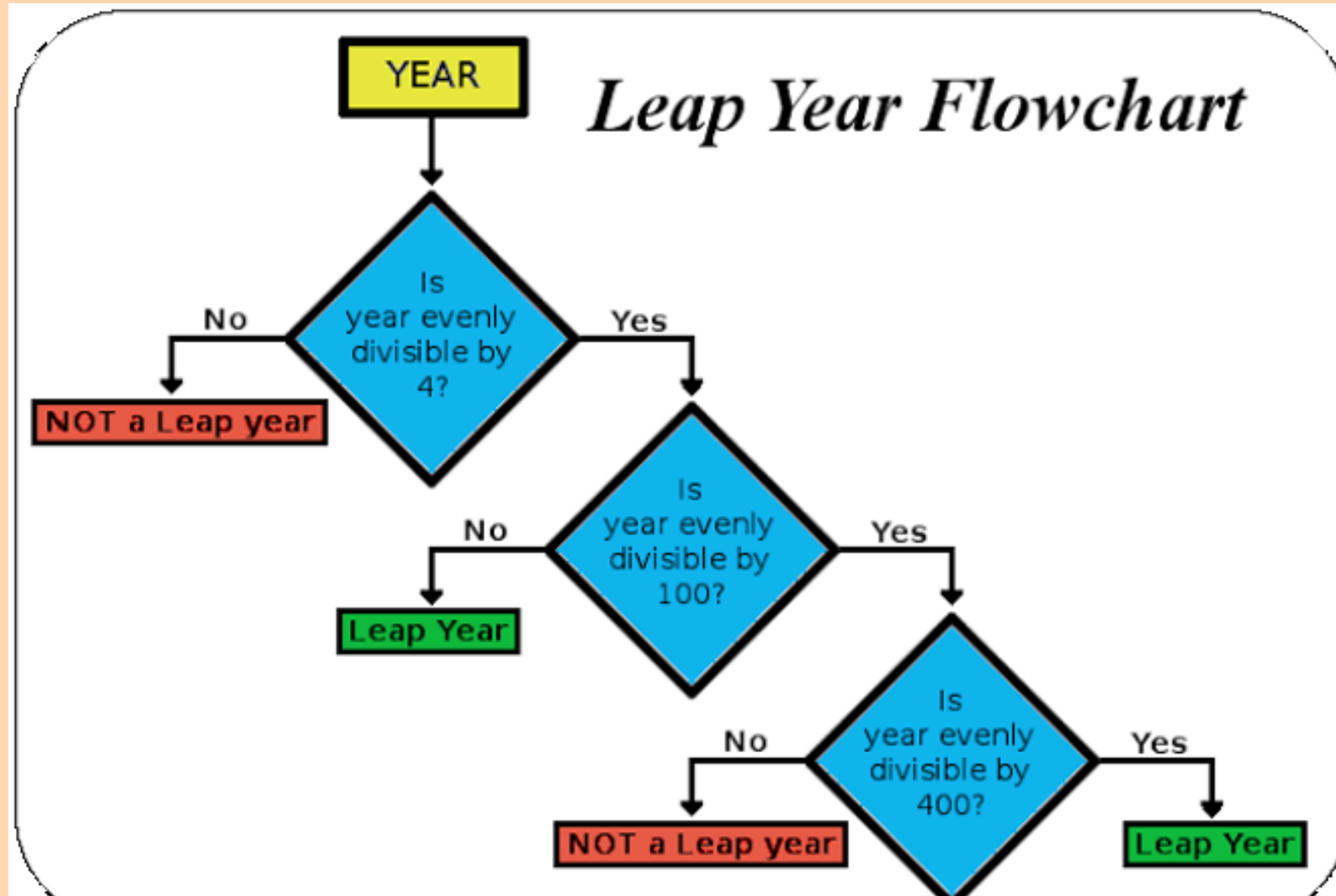


== vs. **=**

```
if (year == 0) { ... }
```

```
if (year = 0) { ... } // ALWAYS FALSE!
```

Leap Year (2): Nested `if/else` (NO)



Leap Year (3): **else if**

```
if (year is not divisible by 4) then (it is a common year)
else if (year is not divisible by 100) then (it is a leap year)
else if (year is not divisible by 400) then (it is a common year)
else (it is a leap year)
```

Leap Year (4): The Ultimate Version

A year is a **leap year** if

- it is divisible by 4 but not by 100,
- except that years divisible by 400 are leap years.

Short-circuit Evaluation (短路求值)

```
// test: year = 25
// test: year = 80
// test: year = 100
// test: year = 400
if ((year % 4 == 0 && year % 100 != 0) ||
    (year % 400 == 0)) {
    leap = 1;
}
```

Order of Evaluation

For most operators (except `&&`, `||`, `?:`),
operand evaluation order is **unspecified**.

```
int x{i + i++};
```



[[likely]] / [[unlikely]] (C++20)

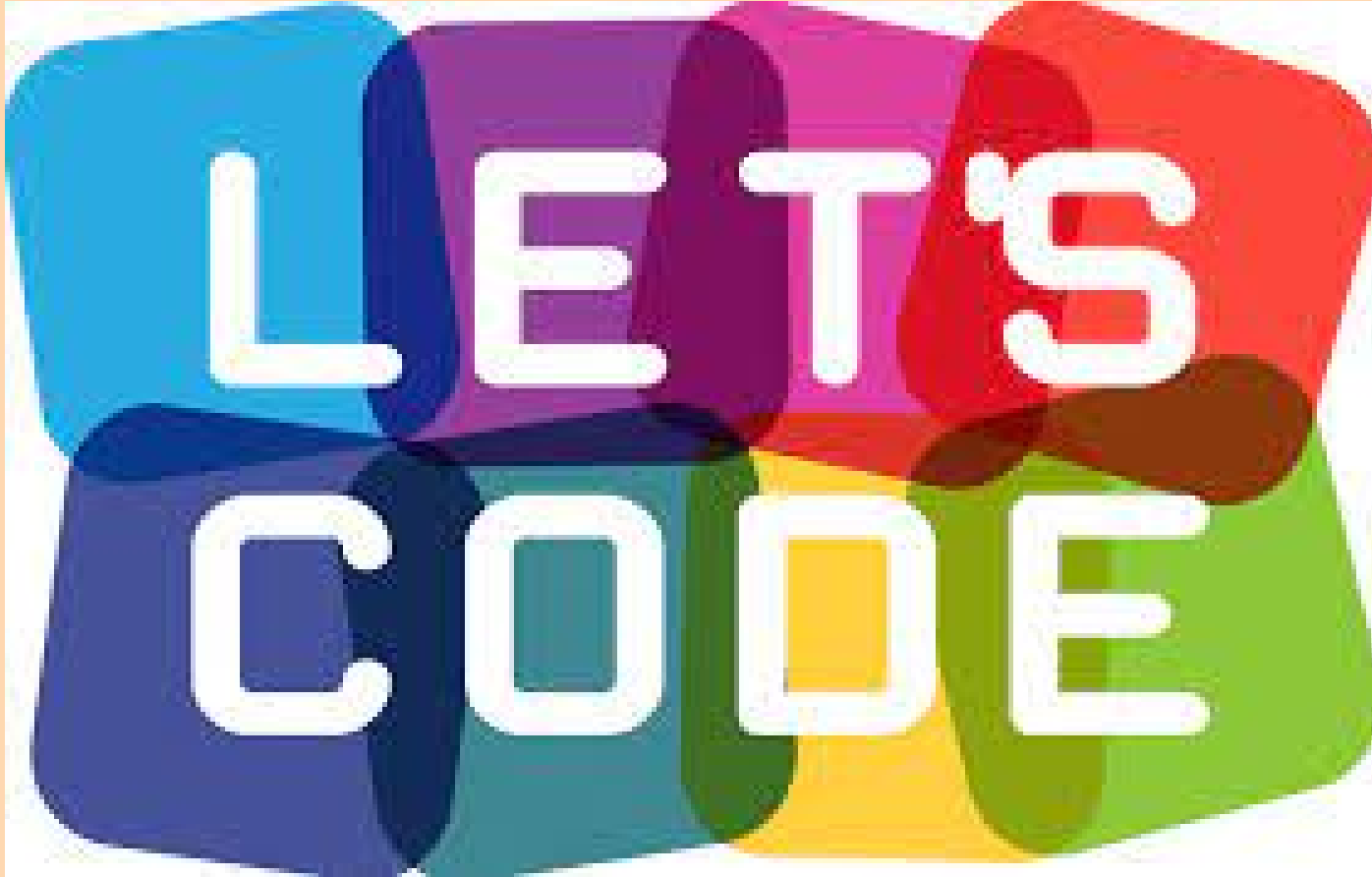
(Only) a hint to compiler for branch prediction

Next Day Calculator

Given (year, month, day), print the next day.

2024-2-28 → 2024-2-29

2024-12-31 → 2025-1-1



days-of-month.cpp temperature-control.cpp

Fall-through & Case Scope

- Missing `break` = implicit fall-through (usually a bug)
- Intentional fall-through: use `[[fallthrough]]`
- Variables in `case`: use `{}` to limit scope

switch with Initializer (C++17)

Temperature Control

if-else vs. switch

Prefer **switch** for discrete constants (CGL ES.70)

